



Manifold for T1D – UBC SC-β Design Analytics

Stem-cell-derived β-cell differentiation, maturation, and graft-performance inference

SC-β cells have crossed from protocol to clinical insulin-independence. What is not yet established at the analytic level required for next-generation program design is the directed causal structure connecting **differentiation-protocol parameters, maturation-state biomarkers, and post-graft functional outcome**. UBC's Life Sciences Institute — with lineage through the **ViaCyte** program — holds a preclinical-to-clinical translation substrate unmatched elsewhere. **Manifold** configures causal discovery on the differentiation-to-outcome record, criticality estimators (GP regression on protocol surfaces, HTE learners on gene-edit variants, change-point on maturation, cross-species translation) feeding a composite fragility score (ΩF), and Pearl do-calculus with interdiction for encapsulation and gene-edit variant prioritization.

The field's next five years will be defined by protocol-to-performance generalization and by next-generation architecture design — a causal-inference problem on the legacy translation record and the high-dimensional design space.

ΩF PILLAR MAPPING – SC-β DESIGN AND GRAFT-PERFORMANCE PHYSIOLOGY

ΩF PILLAR	SC-β READING
I — Interaction	Protocol × maturation × genetic-modification × graft-architecture interactions; SC-β × host-immune compatibility
R — Reserve	Maturation-state functional mass (GSIS, insulin content, expression panel); post-graft functional mass
J — Junction	Immune–metabolic coupling across translation; encapsulation-barrier × host-immune coupling
C — Cascade	Differentiation-maturation cascade (NKX6.1, MAFA, PDX1); dedifferentiation cascades; encapsulation-fibrosis cascades
T — Temporal	Hazard of maturation regression, graft failure, response loss; time-to-functional-maturity across species

PROPOSED PARTNERSHIP

Partnership with a principal investigator at the UBC Life Sciences Institute is sought for:

1. Retrospective differentiation-protocol and preclinical SC-β performance data access under UBC's existing Data Use Agreement.
2. Co-authorship on the protocol-to-performance causal-integration and next-generation-variant interdiction benchmark manuscript (twelve to fourteen weeks from data access).
3. Co-application to Breakthrough T1D Cures, Helmsley Restoring Insulin Production, CIHR, or NIH NIDDK for the Phase 2 translational arm.

Apex Analytica contributes the causal-inference engine, the ΩF framework, and the engineering team. Full brief: M-T1D-02-UBC .